
Does a Transformer Read the Structure of a Graph It Is Conditioned On, or Just Its Presence?

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A Null Result, a Retraction, and What Survives Six Seeds | September 2026

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Abstract

We condition Transformer attention on an explicit dependency graph that each layer iteratively re-estimates, and test whether the graph’s structural *content* matters or whether its mere presence is doing the work. A first run at three seeds per condition appeared to answer this cleanly: a graph containing zero correct arcs outperformed one containing plausible structure, on every seed. At six seeds that result reverses sign and the apparent finding evaporates — we report the retraction explicitly, since it is itself the more instructive result. What survives at six seeds is a single robust, small effect: attention conditioned on *any* graph tensor beats attention conditioned on none, by a consistent but modest margin (5 of 6 seeds, paired $t = 2.22$), while the choice between a structurally-informative graph and an uninformative one of matched density makes no measurable difference. At the 800-sentence scale tested, the mechanism behaves like a generic attention bias rather than a structural prior, and the training regime is heavily overfit throughout, which we argue is a precondition for, not yet an explanation of, that finding.

1 Motivation

A Transformer whose attention is additively biased by an explicit graph, and which re-estimates that graph from its own representations at every layer, is a natural test bed for a specific question: does the model exploit the graph’s structural content, or does it merely benefit from having *some* externally-supplied signal to attend through, regardless of whether that signal is informative? These are different claims with different implications, and they are separable only by comparing an informative graph against an uninformative one of matched density — comparing a graph-conditioned model against an unconditioned one cannot distinguish them.

2 Mechanism

Each attention layer’s logits are additively biased by a symmetrized current graph estimate, $QK^\top / \sqrt{d} + \lambda(G + G^\top)$, with λ learned; after the attention and feed-forward sublayers, a bilinear head re-estimates the graph from the layer’s own updated representations, and that estimate biases the next layer. The initial graph handed to the model is deliberately naive: every token’s head is guessed as the immediately preceding token. The task is unlabelled dependency-arc prediction (unlabelled attachment score, UAS) on a ~ 800 -sentence subset of an annotated treebank, against a matched vanilla Transformer baseline that never receives any graph.

The control. Three initial-graph conditions isolate content from presence: *chain* (the naive linear-chain guess; on a sample sentence, 2 of 28 arcs are correct), *random* (identical arc count and one-head-per-dependent structure, heads drawn uniformly — graph-shaped but uninformative; 0 of 28 correct), and *empty* (no graph at all).

3 A Result That Did Not Survive More Seeds

At three seeds, the comparison across conditions appeared to answer the motivating question directly: the *random* condition beat *chain* by +0.018 paired accuracy, on 3 of 3 seeds. Read at face value, this says the model does not use the graph’s structural content — an uninformative graph works at least as well as an informative one.

Six seeds reverse the sign. Adding three more seeds to each condition flips every comparison (Table 1). The three-seed result was noise that happened to look like a clean, reportable mechanism finding, and it is reported here rather than silently replaced, because the reversal is itself evidence about how many seeds this kind of claim requires.

Condition	Paired Δ (3 seeds)	Paired Δ (6 seeds)
chain	+0.000	−0.007
random	+0.018 (3/3 ahead)	−0.004 (3/6 ahead)
empty	−0.012	−0.014

Table 1: Graph-conditioned minus unconditioned accuracy, by initial-graph condition. The apparent random-beats-chain effect does not survive.

At six seeds, graph conditioning does not beat the unconditioned baseline in *any* condition: every point estimate in Table 1 is negative, and none reaches even 4 of 6 seeds in the favorable direction.

4 What Survives: Presence, Not Content

Comparing conditions directly against each other, rather than each against its own baseline, isolates one effect that does hold up (Figure 1):

Comparison	Paired Δ	Seeds ahead
chain – empty	+0.008 $\pm$ 0.008 ($t = 2.22$)	5/6
random – empty	+0.013 $\pm$ 0.033	3/6
random – chain	+0.005 $\pm$ 0.028	3/6

Table 2: Within-seed comparisons between conditions. Only chain vs. empty is consistent; it is also the tightest-spread comparison in the study.

Supplying some graph tensor beats supplying none, consistently and with the tightest spread of any comparison in this study; the choice between an informative and an uninformative graph of matched density does not reach the same level of support in either direction.

This does not establish that content is irrelevant — the random-vs-chain comparison has roughly four times the spread of the chain-vs-empty effect, so the experiment lacks the power to rule content in or out. It establishes only that the presence effect is larger and more robust than any content effect this design can detect.

5 The Training Regime Is Heavily Overfit

Every condition reaches ~ 0.63 training UAS against ~ 0.38 development UAS by the end of training, with development loss rising over the final third of training while training loss continues to fall. At 800 sentences, both the conditioned and unconditioned models have enough capacity to memorize the training set. This is a plausible precondition for the null result in Table 1 — a model already

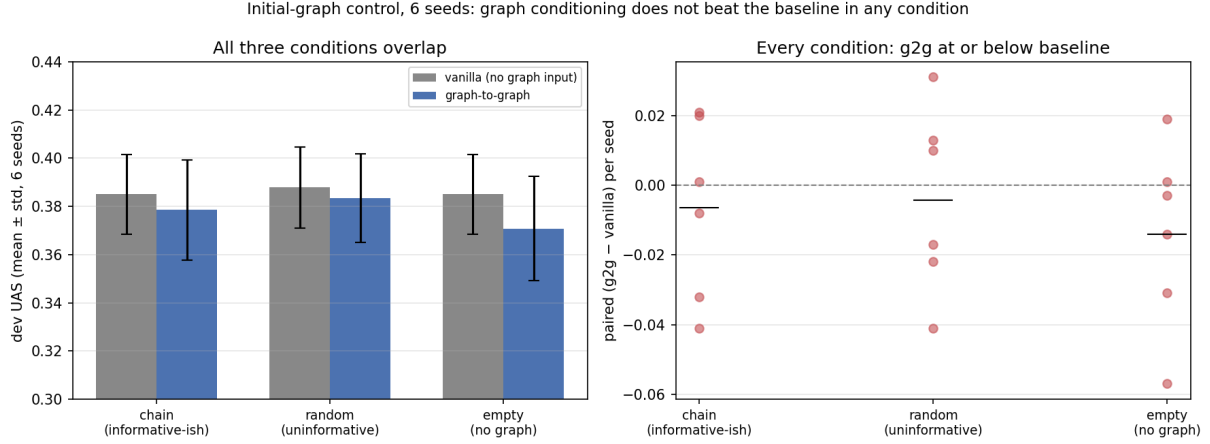


Figure 1: Six-seed result. Left: all three conditions’ accuracy distributions overlap. Right: per-seed paired differences (graph-conditioned minus baseline); every condition sits at or below zero.

fitting noise in its training data has limited room for an additional structural signal to improve generalization — but it is not yet a tested explanation of it; that would require repeating this comparison at a scale where neither model overfits.

6 A Methodological Note

The *random* condition’s own retraining of the graph consumed the global random-number stream during data construction, which perturbed model initialization relative to the other two conditions at the same seed — detectable because *chain* and *empty*, which never touch that stream, produced bit-identical baseline accuracies across all six seeds while *random* did not. A dedicated random-number generator, scoped to graph construction only, resolves this. The discrepancy it caused (0.0007 accuracy) is small relative to the effects in Table 2, but the *random* arm should be re-run under the fix before its numbers are relied on further.

7 Discussion

Two lessons, at different levels. Methodologically, a three-seed comparison here produced a directionally consistent, statistically clean-looking result that six seeds overturned — and it overturned in the specific way that is easy to miss, because the three-seed version was not obviously wrong on its face. Mechanistically, once seed count was adequate, the only effect that survived was the coarsest possible one — graph versus no graph — while the finer question this design was built to answer, whether structural correctness matters, remains statistically open rather than answered in either direction. A negative result reported with the power analysis to say so is more useful than a positive one that later turns out not to replicate.

8 Limitations

- 800 training sentences is a small fraction of the available treebank; every conclusion here concerns this specific, heavily-overfit data regime.
- Six seeds is enough to reverse a three-seed conclusion but not enough for high confidence in what remains: the chain-vs-empty effect’s paired $t = 2.22$ at $df = 5$ corresponds to $p \approx 0.08$,

suggestive rather than conclusive.

- Unlabelled attachment only; a labelled-arc, full-treebank evaluation would be needed before any claim here bears on parsing performance as such.
- The *random* condition’s numbers predate the random-number-generator fix described in Section 6 and should be treated as provisional pending a re-run.

9 Path Forward

The training-scale confound in Section 5 is the most direct next step: repeating the three-condition comparison at increasing training-set sizes would show whether the null result in Table 1 is a property of this architecture or an artifact of a regime where nothing generalizes well enough to show a difference. A second, complementary experiment follows from Section 4’s power limitation: replacing the three discrete conditions with a graded one — true dependency arcs with a controlled fraction corrupted to noise — would trace a full structure-quality curve instead of comparing three points on it, and is better powered to resolve whether content matters at all.

10 Reproducibility

Code, all three initial-graph conditions at six seeds each, and the random-number-generator fix are available at <https://github.com/saqarkpr/graph-conditioned-attention>.